

Symmetries in Physics — Exercise Sheet 6

Symmetry, projectors, and degeneracy

Variant: Questions only

Exercise 1. In Lecture 6's example of a particle hopping on the corners of a triangle, $C_3 = \langle r \rangle$ acts on the orbitals by

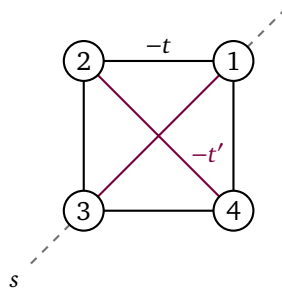
$$\rho(r) |j\rangle = |j+1\rangle,$$

with indices taken modulo 3. This is the regular representation of C_3 (Sheet 3, Exercise 5). The lecture simply wrote down the vectors $v_k = \sum_{j=1}^3 \omega^{-kj} |j\rangle$, where $\omega = e^{2\pi i/3}$. Derive them from the character projectors.

- (i) Write down the three character projectors P_k of C_3 .
- (ii) Apply each of them to $|1\rangle$, and compare the result with v_k .
- (iii) Check that $P_0 + P_1 + P_2 = \mathbb{1}$, and explain why the image of each P_k is a line.

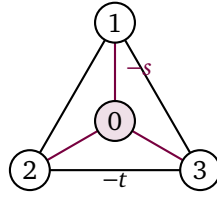
Exercise 2 (retrieval: parity through character projectors). Return to the parity action of $C_2 = \{e, p\}$ on $L^2(\mathbb{R})$ from Sheet 3, Exercise 4: $(\Pi\psi)(x) = \psi(-x)$, where $\Pi = \rho(p)$. Using the two characters and the character-projector formula of Lecture 6, compute both projectors and identify their images. The formula was proved for finite-dimensional spaces, but for a group of order 2 the averaging makes sense on $L^2(\mathbb{R})$ directly, and you may use it in that form. Explain why each image is an isotypic component but is not itself an individual irreducible representation.

Exercise 3. Four identical orbitals $|1\rangle, \dots, |4\rangle$ sit at the corners of a square, numbered anticlockwise. An electron hops between neighbouring corners with amplitude $-t$ and across a diagonal with amplitude $-t'$. The symmetry group is D_4 : the rotation r through $\pi/2$ acts by $|j\rangle \mapsto |j+1\rangle$ (indices modulo 4), and s is the mirror in the diagonal through corners 1 and 3 (dashed).



- (i) With the classes of Lecture 5's D_4 table, $\{e\}$, $\{r^2\}$, $\{r, r^3\}$, $\{s, r^2s\}$, $\{rs, r^3s\}$, show that the permutation character is $\chi_{\text{perm}} = (4, 0, 0, 2, 0)$ and that $\mathbb{C}^4 \cong A_1 \oplus B_1 \oplus E$.
- (ii) Apply the character projectors P_{A_1} , P_{B_1} and P_E to $|1\rangle$.
- (iii) Every irreducible representation occurs once, so the projected vectors are eigenvectors of H . Find the three levels and their degeneracies. For which t' do two levels meet? Is that degeneracy accidental? (Look at H when $t' = t$.)

Exercise 4. Put a fourth atom at the centre of the triangle of Lecture 6, with one orbital $|0\rangle$ of energy ε . The corner orbitals and the hopping $-t$ between corners are as in the lecture, and the hopping between the centre and each corner is $-s$. The group C_{3v} fixes $|0\rangle$ and permutes the corners.



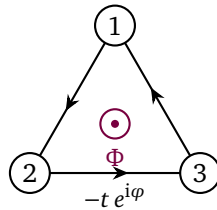
- (i) Show that $\mathbb{C}^4 \cong 2A_1 \oplus E$.
- (ii) Take the orthonormal symmetry-adapted basis $w^{(1)} = |0\rangle$, $w^{(2)} = \frac{1}{\sqrt{3}}(|1\rangle + |2\rangle + |3\rangle)$ of the isotypic component of A_1 . Show that the reduced matrices are

$$F^{(A_1)} = \begin{pmatrix} \varepsilon & -\sqrt{3}s \\ -\sqrt{3}s & -2t \end{pmatrix}, \quad F^{(E)} = t.$$

- (iii) Find the spectrum. Which degeneracy is enforced? For which s is there an accidental threefold level?

Exercise 5. Thread a magnetic flux through the triangle of Lecture 6. Its only effect on the hopping is a phase: an electron hopping anticlockwise, $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$, picks up $e^{i\varphi}$, and one hopping clockwise $e^{-i\varphi}$. With $C|j\rangle = |j+1\rangle$,

$$H = -t(e^{i\varphi} C + e^{-i\varphi} C^{-1}).$$



- (i) Show that H is Hermitian and commutes with $\rho(r) = C$, but that for $\sin \varphi \neq 0$ it commutes neither with the mirror σ of Lecture 6, which fixes corner 3 and swaps 1 and 2, nor with complex conjugation.
- (ii) Show that Lecture 6's vectors $v_k = \sum_j \omega^{-kj} |j\rangle$ are still eigenvectors, with $E_k = -2t \cos(\varphi + \frac{2\pi k}{3})$.
- (iii) Show that two levels coincide exactly when 3φ is a multiple of π . What has happened to the doublet E for other φ , and which of Lecture 6's two reasons for it has the flux removed?